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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 6

Construct a C program to implement preemptive priority scheduling algorithm

PROGRAM:

```
#include <stdio.h>


int main()
{
    int n, i;
    int at[10], bt[10], pr[10], rem[10];
    int wt[10], tat[10];
    int time = 0, completed = 0;
    int pos, highest;

    float avg_wt = 0, avg_tat = 0;

    printf("Enter the number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
```

```
{
    printf("\nEnter details for Process P%d\n", i + 1);

    printf("Arrival Time: ");
    scanf("%d", &at[i]);

    printf("Burst Time: ");
    scanf("%d", &bt[i]);

    printf("Priority: ");
    scanf("%d", &pr[i]);

    rem[i] = bt[i];
}

while(completed < n)
{
    pos = -1;
    highest = 9999;

    /* Find highest priority process */
    for(i = 0; i < n; i++)
    {
        if(at[i] <= time &&
           rem[i] > 0 &&
           pr[i] < highest)
```

```
    {
        highest = pr[i];
        pos = i;
    }
}

/* If no process has arrived */
if(pos == -1)
{
    time++;
}
else
{
    /* Execute for one unit of time */
    rem[pos]--;
    time++;

    /* Process completed */
    if(rem[pos] == 0)
    {
        completed++;

        tat[pos] = time - at[pos];
        wt[pos] = tat[pos] - bt[pos];
    }
}
```

```
    }

    printf("\nProcess\tAT\tBT\tPriority\tWT\tTAT\n");

    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t\t%d\t%d\n",
               i + 1,
               at[i],
               bt[i],
               pr[i],
               wt[i],
               tat[i]);

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt = avg_wt / n;
    avg_tat = avg_tat / n;

    printf("\nAverage Waiting Time = %.2f", avg_wt);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

    return 0;
}
```

OUTPUT:

```
Enter the number of processes: 4

Enter details for Process P1
Arrival Time: 2
Burst Time: 6
Priority: 3

Enter details for Process P2
Arrival Time: 6
Burst Time: 8
Priority: 3

Enter details for Process P3
Arrival Time: 8
Burst Time: 4
Priority: 1

Enter details for Process P4
Arrival Time: 9
Burst Time: 4
Priority: 0

Process AT      BT      Priority        WT      TAT
P1      2       6       3               0       6
P2      6       8       3               10      18
P3      8       4       1               4       8
P4      9       4       0               0       4

Average Waiting Time = 3.50
Average Turnaround Time = 9.00

--------------------------------
Process exited after 24.26 seconds with return value 0
Press any key to continue . . .
```